

sources_1\new\INPUT_SYNC.vhd

```
1  library IEEE;
2  use IEEE.STD_LOGIC_1164.ALL;
3
4  --*****
5  --*
6  --* Name: INPUT_SYNC
7  --* Designer: Matthew Mwangi
8  --*
9  --*      This is a component that handles synching its input in this case the aSyncIn
10 --*      active clock, to avoid metastability issues.
11 --*
12 --*****
13
14
15 entity INPUT_SYNC is
16   Port (
17     reset: in std_logic;
18     clock: in std_logic;
19     aSyncIn: in std_logic;
20     syncOut: out std_logic
21   );
22 end INPUT_SYNC;
23
24 architecture INPUT_SYNC_ARCH of INPUT_SYNC is
25   constant ACTIVE: std_logic := '1';
26   signal syncChain: std_logic_vector(1 downto 0);
27 begin
28   process(reset, clock)
29   begin
30     if reset = ACTIVE then
31       syncChain <= (others=>'0');
32     elsif rising_edge(clock) then
33       syncChain <= syncChain(0) & aSyncIn;
34     end if;
35   end process;
36
37   syncOut <= syncChain(1);
38 end INPUT_SYNC_ARCH;
```

